

# Structural Properties of Null Gene Regulatory Networks

Andy Halleran and John Marken

## Introduction

In 2014, Albergante *et al.* published a paper in *eLife* entitled “Buffered Qualitative Stability explains the robustness and evolvability of transcriptional networks” [Albergante *et al.*, 2014]. This paper put forth the concept of Buffered Qualitative Stability (BQS) as a fundamental topological property of healthy gene regulatory networks (GRNs) that is robust to the random addition and deletion of nodes and edges. The BQS property itself is mainly characterized by two measures:

- (i) An absence of long feedback loops, and
- (ii) An absence of nodes with both a high in-degree and a high out-degree.

The authors counted the number of complete and incomplete feedback loops in published GRN datasets. When compared with null network models, they found that real-world networks exhibited a striking lack of such loops—this was the basis for their claim that BQS is a property of real-world GRNs.

However, these results at first glance seemed contradictory with published results from Uri Alon’s lab, which claimed that (i) there do exist long feedback loops in the *E. coli* GRN [Shen-orr *et al.*, 2002], and (ii) topological network properties like certain structural motifs are not significantly overexpressed in real-world GRNs compared to null network GRNs [Milo *et al.*, 2002].

For our project, we wanted to determine what was causing the dissonance in interpretation between the two papers.

## Null Models

We first noticed that the algorithms used to generate null GRNs were quite different between the BQS and the Alon papers. The BQS paper used a number of different null models, all of which were grounded in the same notion of randomly adding edges to a specified set of nodes. Meanwhile, the Alon null took a real-world GRN and simply reshuffled the connections between the nodes, preserving the in- and out-degree on each node. Thus we felt the Alon Null adhered strictly to the structural properties of a real GRN, whereas the BQS Null disposed of too many. In order to determine if consonance could be found between the papers by using null models that were not as extreme on the strictness scale, we came up with three additional null network algorithms. Thus we had five null network models to work with:

- BQS Null:
  - Input: number of Nodes, number of Transcription Factors (TFs), number of Edges
  - Algorithm:
    - 1. For each non-TF node (Leaf), randomly choose a TF and connect an edge from that TF to the Leaf.
    - 2. Randomly add edges between TFs and Nodes until there are the correct number of Edges in the graph.
- Marginals Null:

- Input: The in- and out-degree marginal distributions from a real GRN, the number of nodes
- Algorithm:
  - 1. For each node, sample an in-degree from the marginal in-degree distribution and an out-degree from the marginal out-degree distribution.
  - 2. Randomly connect the node's edges to other nodes in such a way that other nodes' degree distributions, should they already be assigned, are not modified.
- Correlated Marginals Null:
  - Input: The in- and out-degree marginal distributions from a real GRN, the correlation coefficient between them, and the number of nodes
  - Algorithm:
    - 1. For each node, sample an in- or out-degree from the correct marginal distribution, then use the correlation coefficient to calculate the other directed degree for the node.
    - 2. Randomly connect the node's edges to other nodes in such a way that other nodes' degree distributions, should they already be assigned, are not modified.
- Bivariate Null:
  - Input: The bivariate degree distributions from a real GRN, the number of nodes
  - Algorithm:
    - 1. For each node, sample an in- and out-degree from the bivariate degree distribution.
    - 2. Randomly connect the node's edges to other nodes in such a way that other nodes' degree distributions, should they already be assigned, are not modified.
- Alon Null:
  - Input: A real GRN
  - Algorithm:
    - 1. Disconnect all of the edges in the network, but preserve the in- and out-degree of each node.
    - 2. Randomly reconnect all stubs on each node.
  - Equivalent Algorithm:
    - 1. Choose two edges (A->B) and (C->D) and shuffle them such that they become (A->D) and (C->B).
    - 2. Repeat step 1 a sufficient number of times

Another distinction that we noted between the BQS and Alon papers was that they used two different networks as their 'true' *E. coli* GRN. Because the Alon paper was published almost 15 years before the BQS paper, they used an earlier version of RegulonDB that consisted of 423 nodes, 116 TFs, and 578 Edges. Meanwhile, the BQS authors used a more recent version of RegulonDB that consisted of 1827 nodes, 199 TFs, and 4348 Edges.

Thus we arrived at our main research question for our project: To what extent do (i) the choice of null network model and (ii) the features of the input GRN affect the interpretation of any comparisons about structural network properties and functions?

## Results

### *Different Null Models display Different Network Properties*

In order to best assess the performance of a null network model, we decided to first replicate the results obtained in the original BQS paper. Here they counted the number of complete and incomplete feedback loops in the network, and showed that the null models exhibited an exponentially increasing number of loops as their length increased. Additionally, we measured two properties of whole-network connectivity at the vertex level: betweenness and eigenvector centrality.

We found that the Alon, Marginal, and Bivariate Null Networks were able to reproduce the qualitative shape of the Complete Feedback Loop distribution from the BQS paper when applied to the *E. coli* GRN used by the Alon paper (Small Coli), but that the BQS Null Model was unable to do so [Fig. 1].

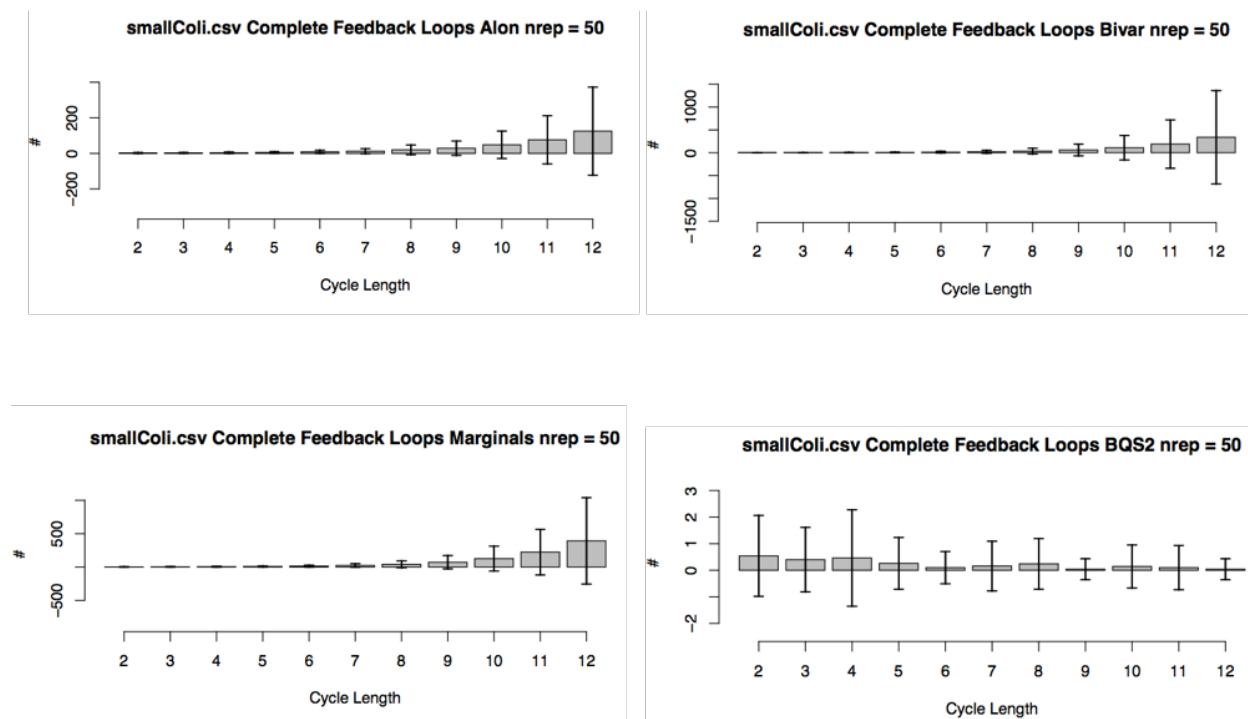


Figure 1: Complete Feedback Loops in the *E. coli* GRN from the Alon paper

We found the same case when counting the number of Incomplete Feedback Loops in the Null Networks [Fig. 2].

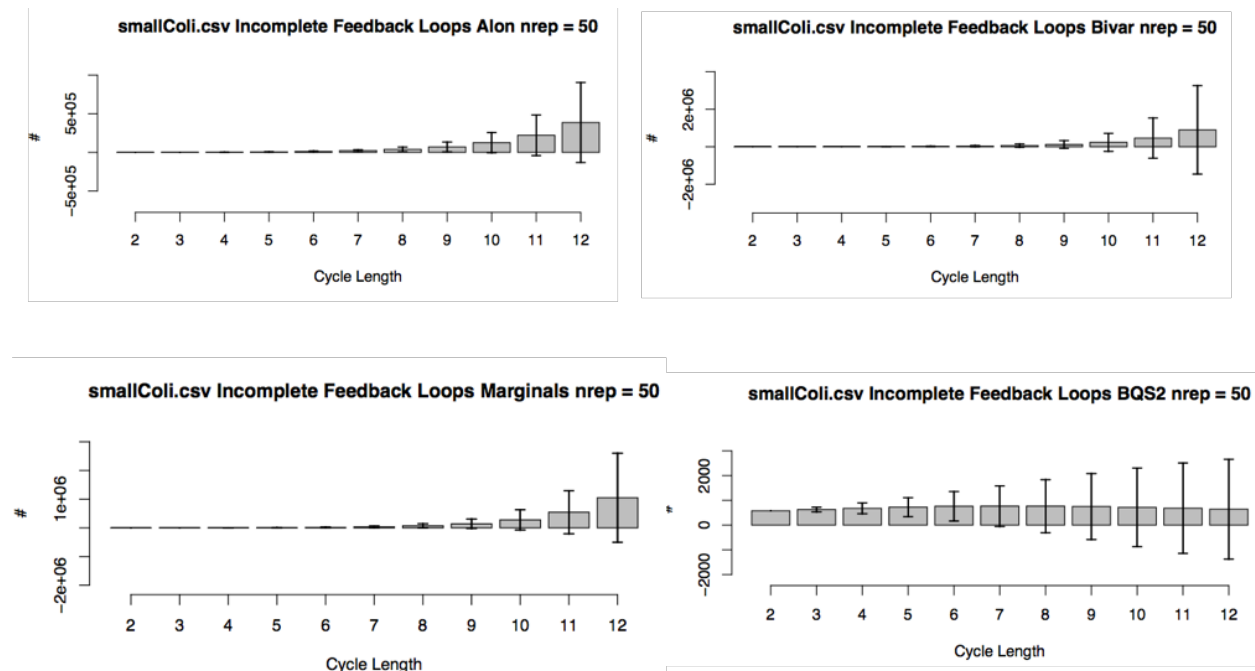


Figure 2: Incomplete Feedback Loops in the *E. coli* GRN from the Alon Paper

We also found that the Alon, Marginal, and Bivariate null models exhibited qualitatively similar distributions for eigenvector centrality, whereas the BQS null model displayed a different one [Fig. 3].

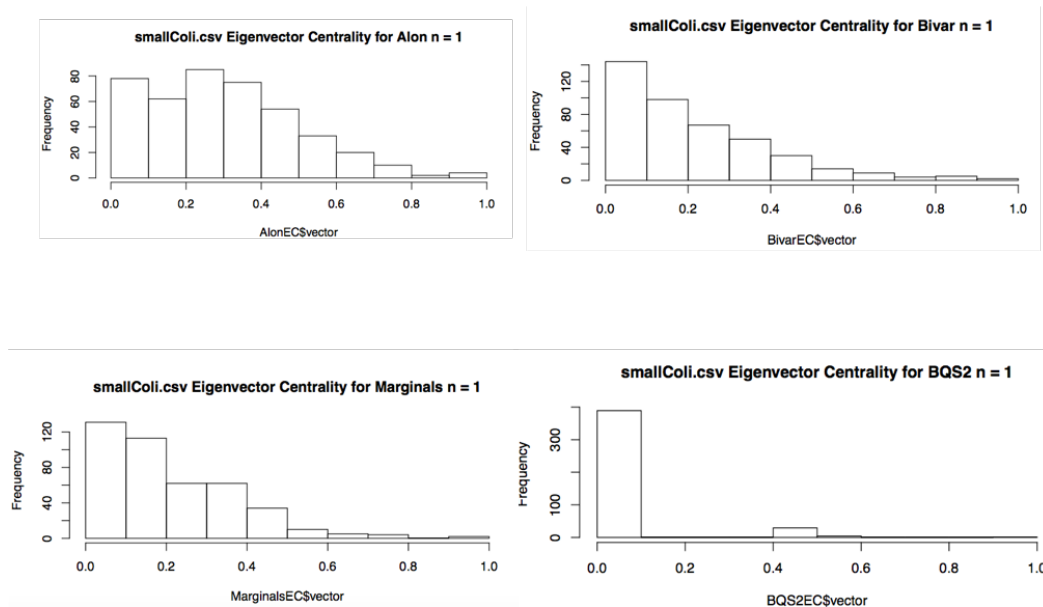


Figure 3: Eigenvector Centrality in the *E. coli* GRN from the Alon Paper

However, betweenness centrality looked fairly consistent across all four null models [Fig. 4].

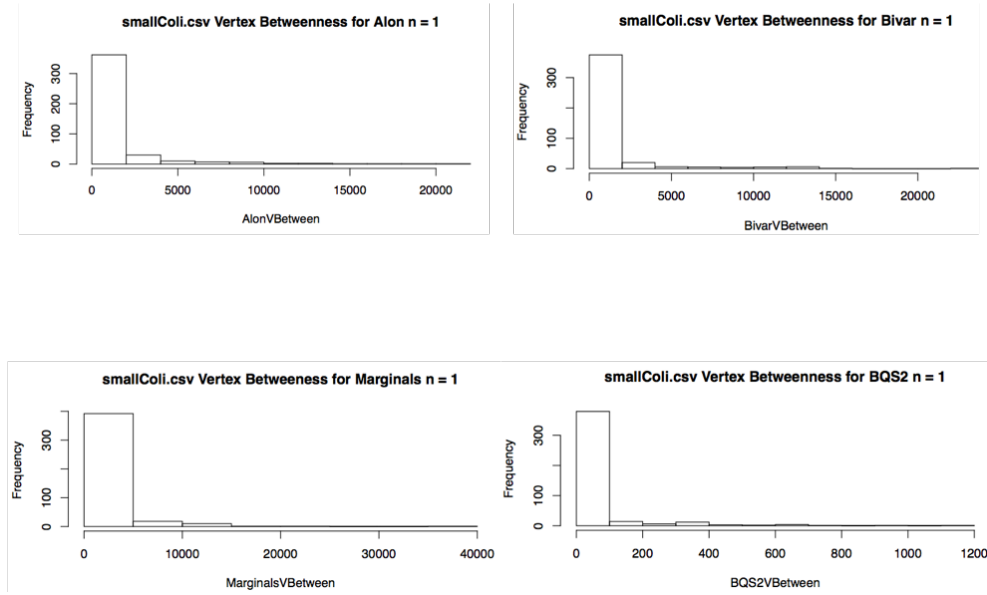


Figure 4: *Betweenness Centrality in the E. coli GRN from the Alon Paper*

The fact that we were unable to reproduce the BQS paper's results using their own null model on a different input GRN made us question the impact of the parameters in the input GRN. We isolated this result to the effect of the GRN, as when we use the same GRN as the BQS paper (regulon Coli), we do not find these discrepancies between the null models and between our results and the paper's [Figs. 5-8].

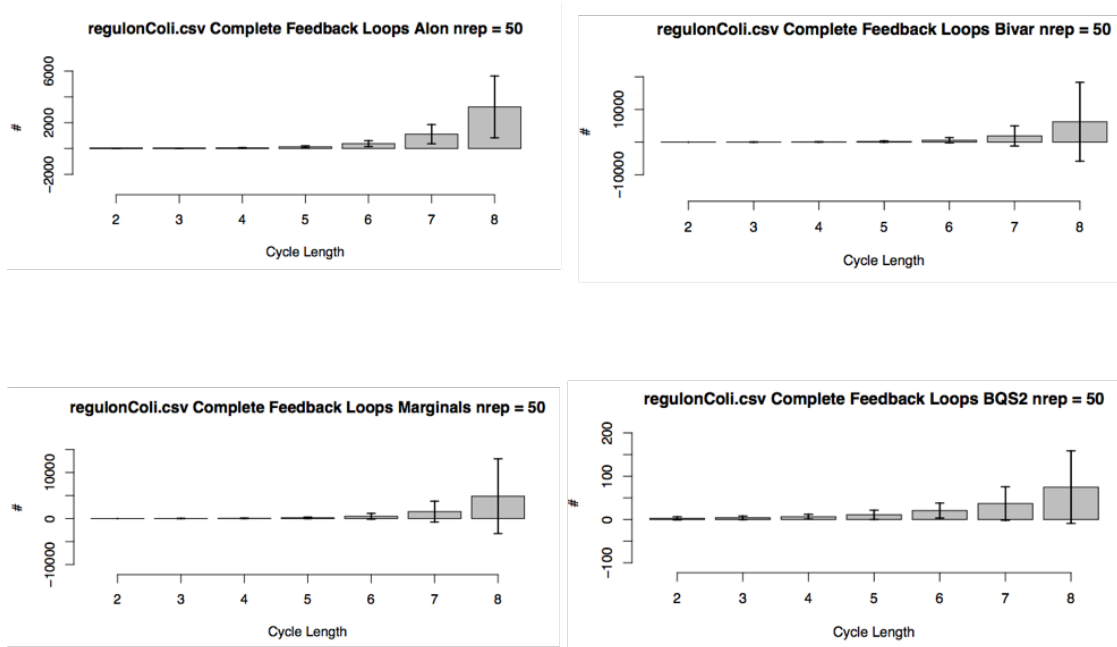


Figure 5: *Complete Feedback Loops for the E. coli GRN from the BQS Paper*

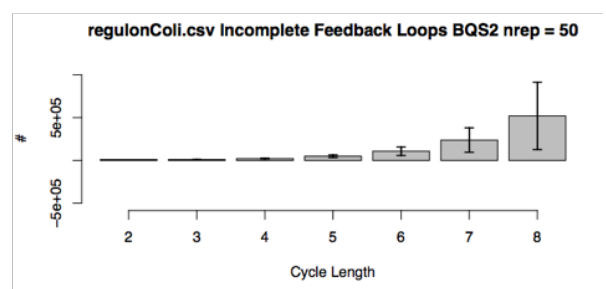
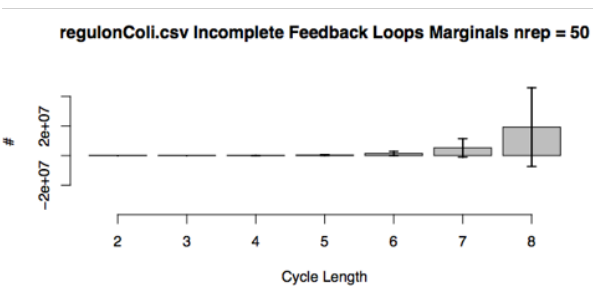
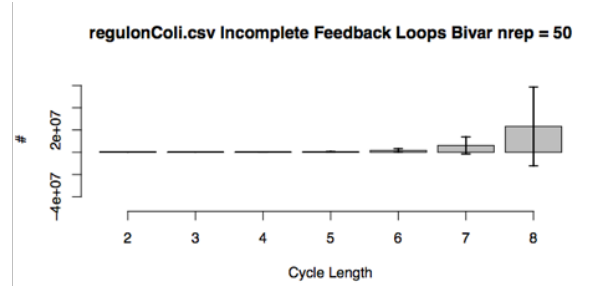
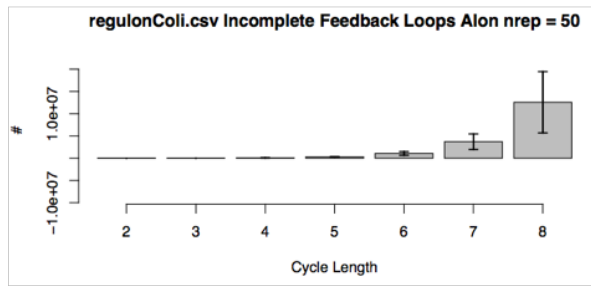


Figure 6: Incomplete Feedback Loops for the *E. coli* GRN from the BQS Paper

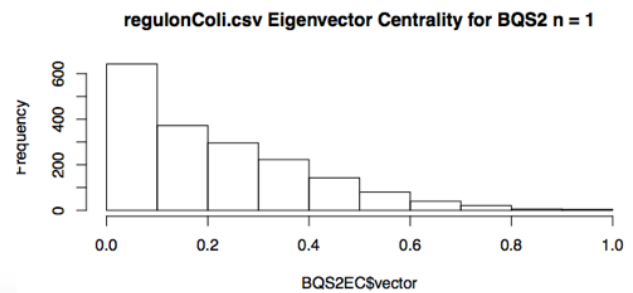
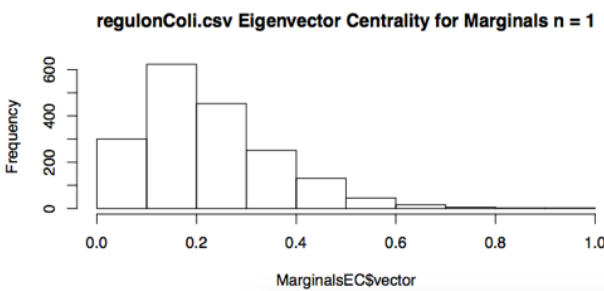
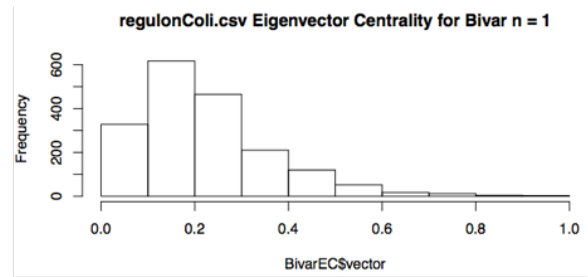
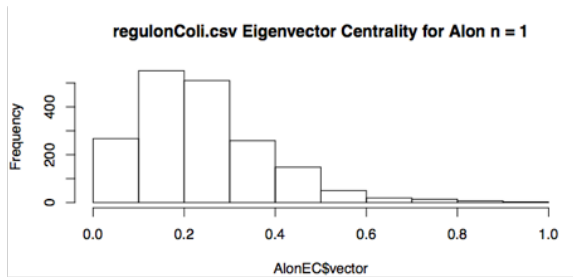


Figure 7: Eigenvector Centrality for the *E. coli* GRN from the BQS Paper

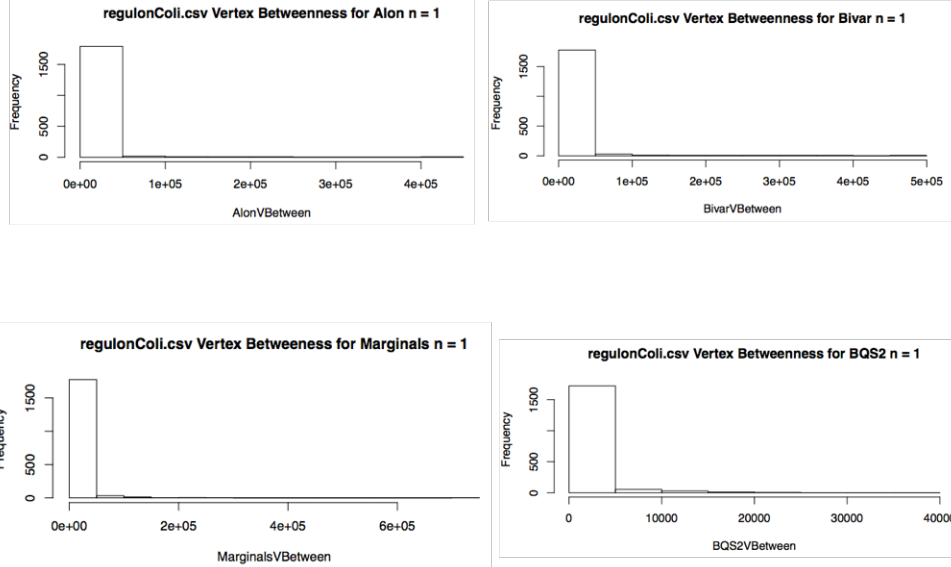


Figure 8: Betweenness Centrality for the *E. coli* GRN from the BQS Paper

### Different Network Parameters yield Different Null Networks

Because the most immediate difference between the two input GRNs (small *Coli* from the Alon paper and regulon *Coli* from the BQS paper) is their size, we wanted to isolate its impact by modifying the other relevant parameters while preserving the size. To do this we took the Regulon *Coli* GRN and applied two transformations: (i) Preserving the TF:Nodes and Edge:Nodes ratios while changing the number of nodes to match Small *Coli*, and (ii) Preserving the number of nodes but changing the TF:Nodes and Edge:Nodes ratios to match those of Small *Coli*. We found that across all our network measures except for betweenness centrality, transformation (i) remained qualitatively similar to the untransformed GRN whereas transformation (ii) did not [Figs. 9-12].

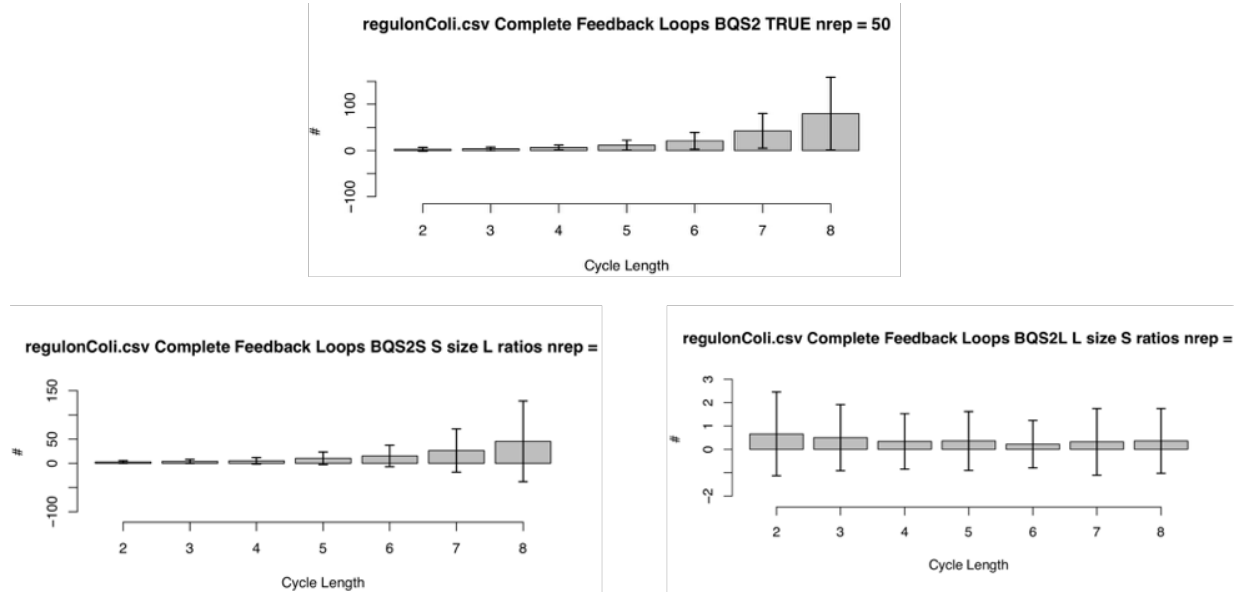


Figure 9: Complete Feedback Loops for (top) the *E. coli* GRN from the BQS Paper, (left) the GRN with ratios preserved but size modified, and (right) the GRN with size preserved but ratios modified

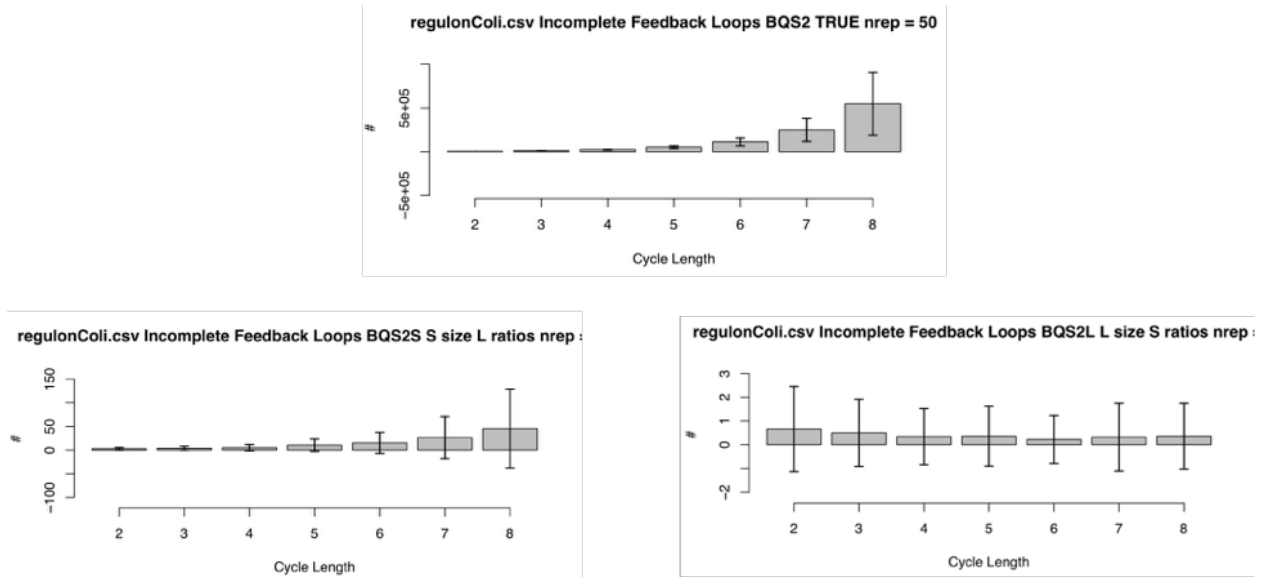


Figure 10: Incomplete Feedback Loops for (top) the E. coli GRN from the BQS Paper, (left) the GRN with ratios preserved but size modified, and (right) the GRN with size preserved but ratios modified

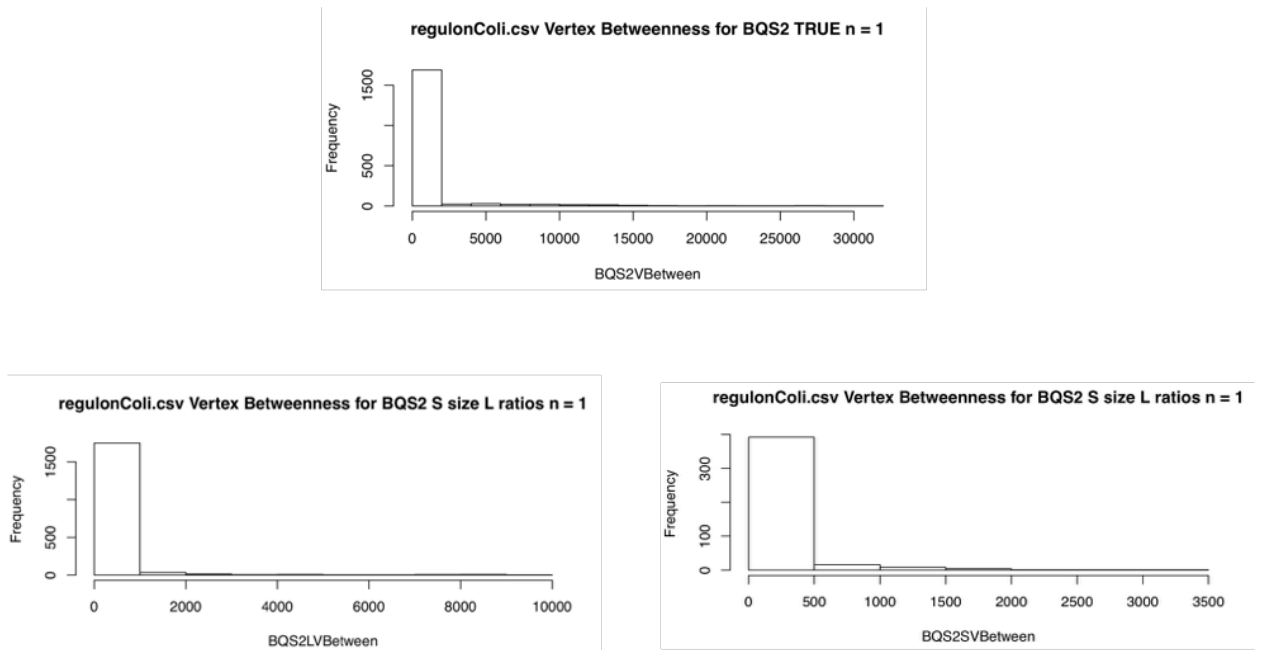


Figure 11: Betweenness Centrality for (top) the E. coli GRN from the BQS Paper, (left) the GRN with ratios preserved but size modified, and (right) the GRN with size preserved but ratios modified



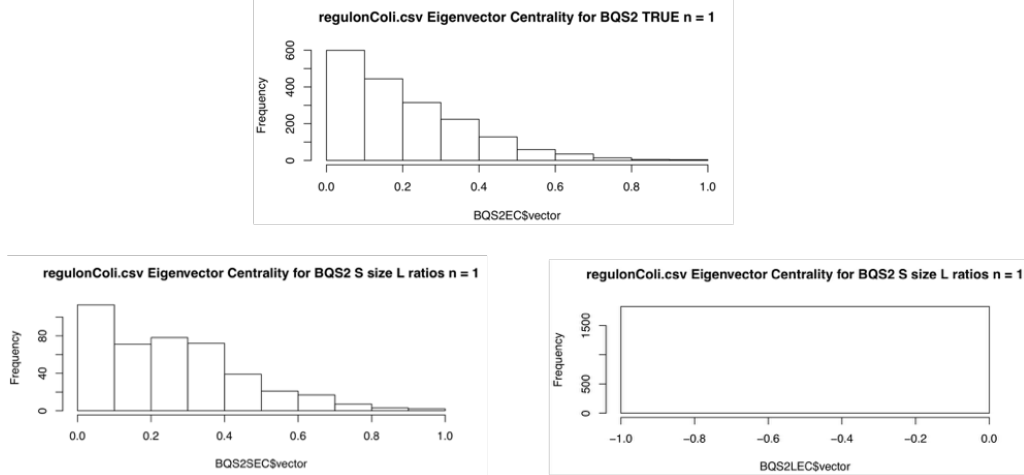


Figure 12: Eigenvector Centrality for (top) the *E. coli* GRN from the BQS Paper, (left) the GRN with ratios preserved but size modified, and (right) the GRN with size preserved but ratios modified

Thus we concluded that the size of the GRN does not drive the difference in performance of the various null models we tested. Instead, the structural properties of the network itself seem to be the important feature in this regard. In order to further pursue this hypothesis, we decided to focus on two specific parameters (the TF:Node ratio and the Edge:Node ratio) and systematically modify them in the BQS Null graph to determine their impact on the resultant null network. Measuring the sum of the Betweenness and Eigenvector Centrality distributions for a given parameter combination, we found that there is a striking relationship between these network parameters and the performance of the null model as assessed by these metrics [Fig. 13].

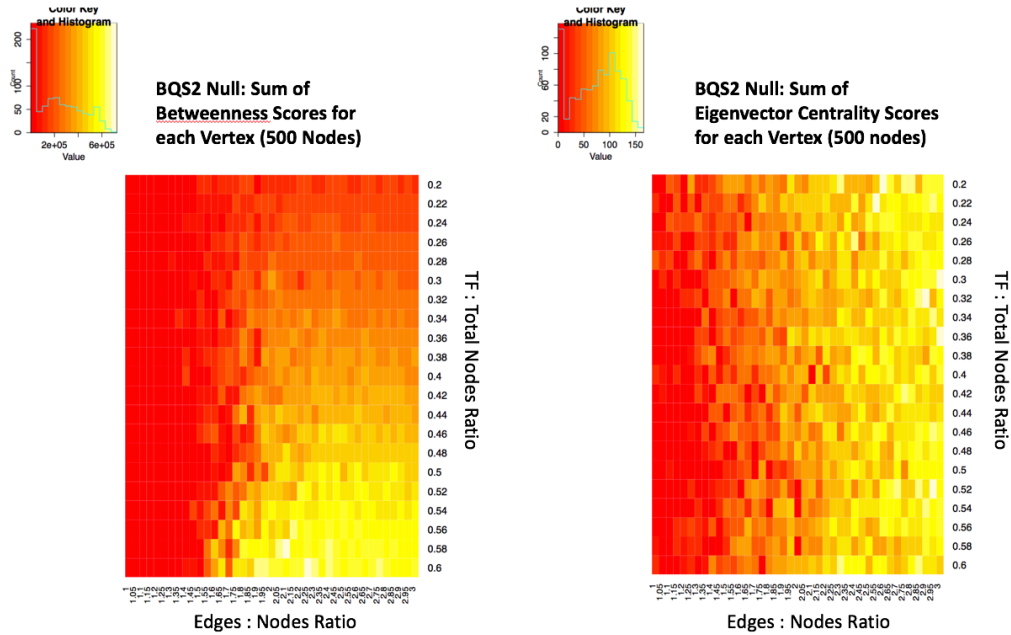


Figure 13: Performance of the BQS Null Network as measured by (left) the sum of the betweenness centrality scores for each vertex and (right) the sum of the eigenvector centrality scores for each vertex, as a function of TF:Nodes and Edges:Nodes ratios. 500 vertices per graph.

It seems that the Betweenness Centrality is affected by a combination of these two parameters, while the Eigenvector Centrality is dependent only on the Edges:Nodes ratio.

We replicated our initial size-invariance result by noting that the graphs look qualitatively identical after applying a ten-fold increase in the number of nodes in the null network [Fig. 14].

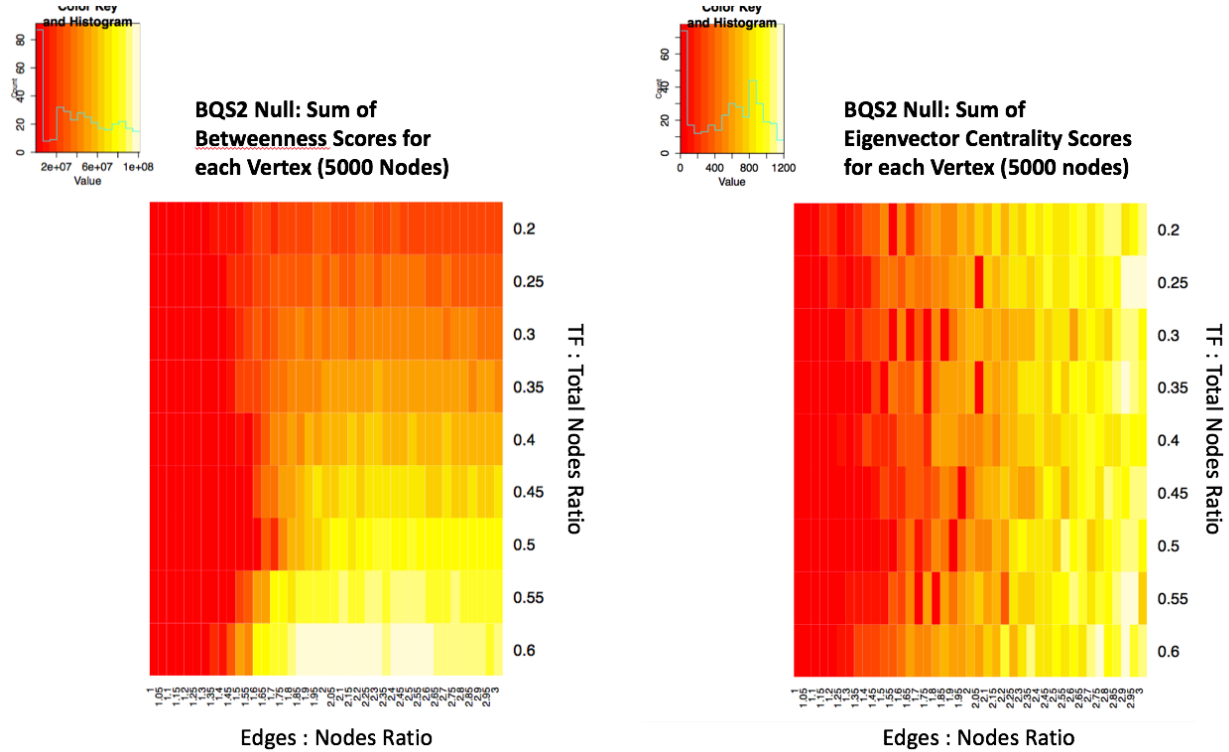


Figure 14: Performance of the BQS Null Network as measured by (left) the sum of the betweenness centrality scores for each vertex and (right) the sum of the eigenvector centrality scores for each vertex, as a function of TF:Nodes and Edges:Nodes ratios. 5,000 vertices per graph.

## Discussion

Our results suggest that both (i) the null model used for comparison to the actual GRN, and (ii) the input parameters into the null model can qualitatively change how you interpret the results of GRN comparisons. However, there is still much work left to be done on this project to make a complete story.

1. We need to look at the number of connected components in the different simulations and see if there is a relationship between # connected components and measures of network structure. This will inform whether future experimenters should use null models that maintain the number of connected components that were input or if it is okay to use methods that do not have that additional constraint
2. We still have not performed simulations with our medium-strictness null model: “correlated marginal null.” While the critical change we observed seems to be happening

- at the jump between marginal null to BQS null, and not from bivariate to marginal, we do not expect any changes to our interpretations when we analyze this null.
3. We need to develop a method for tuning the edges:nodes ratio and the percent TF for other null models. The BQS null was the easiest because there are only three input parameters (numEdges, numVertices, numTFs) that are easily tunable – changing these parameters for the other null graphs will require more thought.
  4. We need to use high performance computing to get finer granularity and a larger sample size for our parameter space plots.

#### References:

1. Albergante, Luca, J. Julian Blow, and Timothy J. Newman. "Buffered Qualitative Stability explains the robustness and evolvability of transcriptional networks." *Elife* 3 (2014): e02863.
2. Shen-Orr, Shai S., et al. "Network motifs in the transcriptional regulation network of Escherichia coli." *Nature genetics* 31.1 (2002): 64-68.
3. Milo, Ron, et al. "Network motifs: simple building blocks of complex networks." *Science* 298.5594 (2002): 824-827.